

MH1300 FOUNDATIONS OF MATHEMATICS

Tutorial 6 Solutions

Ex. 4.3.16. Prove the statements directly from the definition of divisibility.

For all integers a, b , and c , if $a|b$ and $a|c$ then $a|(b - c)$.

SOLUTION . Suppose a, b , and c are integers and $a|b$ and $a|c$. [We must show that $a|(b - c)$.]

By definition of divisibility, there exist integers r and s such that $b = ar$ and $c = as$. Then

$$b - c = ar - as = a(r - s)$$

by substitution and the distributive law. But $r - s$ is an integer since it is a difference of two integers. Therefore $a|(b - c)$ [as was to be shown]. $\square$

Ex. 4.3.x. Prove the following statements if it is true, and give a counterexample if it is false:

(20) The sum of any three consecutive integers is divisible by 3.

(21) The product of any two even integers is a multiple of 4.

(23) A sufficient condition for an integer to be divisible by 8 is that it be divisible by 16.

(25) For all integers a, b, c , if a divides c then ab divides c .

(30) For all integers a, n , if $a | n^2$ and $1 \leq a \leq n$, then $a | n$.

SOLUTION . (20): This is true. Let n be the smallest of the three integers. Then, the three integers are $n, n + 1$ and $n + 2$. The sum of the three integers is $n + (n + 1) + (n + 2) = 3n + 3 = 3(n + 1)$. Since $n + 1 \in \mathbb{Z}$, this means that the sum of $n, n + 1$ and $n + 2$ is divisible by 3.

(21): This is true. Let n and m be two arbitrary integers that are both even. (WTS: $4 | nm$). Since n and m are even, let k, l be integers such that $n = 2k$ and $m = 2l$. Then $nm = (2k)(2l) = 4(kl)$. Since $kl \in \mathbb{Z}$, this means that nm is divisible by 4.

(23): “A sufficient condition for an integer to be divisible by 8 is that it be divisible by 16” is translated as “If an integer is divisible by 16, then it is divisible by 8”. This is of course true. Let n be an integer that is divisible by 16. Then there is an integer k such that $16k = n$. Then, $n = 16k = 8(2k)$. Since $2k \in \mathbb{Z}$, this means that n is divisible by 8.

(25): This is false. The statement is “ $\forall a, b, c \in \mathbb{Z}, a \text{ divides } c \rightarrow ab \text{ divides } c$ ”. We need to show the negation is false, that is, we need to show the statement “ $\exists a, b, c$ such that a divides c and ab does not divide c ” is true. There are many choices for a, b, c . For example, take $a = 1, b = 2, c = 1$. Then $a | c$ is true since $1 | 1$ is true. But $ab = 2$ and does not divide $c = 1$.

(30): This is false. We need to find integers a, n such that $a | n^2$ and $1 \leq a \leq n$ and $a \nmid n$. For example, take $a = 4$ and $n = 6$. Then $a = 4$ divides $n^2 = 36$, and $1 \leq a \leq n$ is also true. Furthermore, $a = 4$ does not divide $n = 6$. $\square$

Ex. 4.3.26. Prove the statement directly from the definition if it is true, and give a counterexample if it is false.

For all integers a, b , and c , if $ab|c$ then $a|c$ and $b|c$.

SOLUTION . Suppose a, b , and c are any integers such that $ab|c$.

By definition of divisibility, $c = r(ab)$ for some integer r . Regrouping shows that

$$c = (ra)b \quad \text{and} \quad c = (rb)a.$$

Now both ra and rb are integers because they are products of integers. Thus

$$c = (\text{an integer}) \cdot b \quad \text{and} \quad c = (\text{an integer}) \cdot a.$$

It follows by definition of divisibility that $a|c$ and $b|c$. □

Ex. 4.3.39. Suppose that in standard factored form $a = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}$, where k is a positive integer; $p_1, p_2, \dots, p_k$ are prime numbers; and $e_1, e_2, \dots, e_k$ are positive integers.

a. What is the standard factored form for a^3 ?

b. Find the least positive integer k such that $2^4 \cdot 3^5 \cdot 7 \cdot 11^2 \cdot k$ is a perfect cube (i.e., equals an integer to the third power). Write the resulting product as a perfect cube.

SOLUTION . a. $p_1^{3e_1} \cdot p_2^{3e_2} \cdots p_k^{3e_k}$.

b. $k = 2^2 \cdot 3 \cdot 7^2 \cdot 11$

$$\text{product} = 2^4 \cdot 3^5 \cdot 7 \cdot 11^2 \cdot k = 2^6 \cdot 3^6 \cdot 7^3 \cdot 11^3 = (2^2 \cdot 3^2 \cdot 7 \cdot 11)^3 = 2772^3.$$

□

Ex. 4.3.48. Prove that for any nonnegative integer n , if the sum of the digits of n is divisible by 3, then n is divisible by 3.

SOLUTION . Let n be a nonnegative integer, such that the sum of the digits of n is divisible by 3. We write $n = d_k d_{k-1} d_{k-2} \cdots d_1 d_0$ where $d_k, \dots, d_0$ are the digits of n . We assume that the sum of the digits of n is divisible by 3, in other words, there is an integer l such that $3l = d_0 + d_1 + \cdots + d_{k-1} + d_k$. We can obtain the numerical value of the integer n by expressing it in base 10, and the digits $d_0, \dots, d_k$ of n are just the coefficients of the respective powers of 10.

For example, the number 234 is just $200 + 30 + 4 = 2 \cdot 10^2 + 3 \cdot 10 + 4$. Similarly, we may express $n = d_k \cdot 10^k + d_{k-1} \cdot 10^{k-1} + \cdots + d_1 \cdot 10 + d_0$. We need to show that n is divisible by 3. Notice that for every j , the sum of the GP (geometric progression) $9 + 9 \cdot 10 + 9 \cdot 10^2 + \cdots + 9 \cdot 10^j = 10^{j+1} - 1$. Therefore, $10^{j+1} - 1$ is divisible by 3 for every $j \geq 0$. Thus, we can write:

$$\begin{aligned} n &= d_k \cdot 10^k + d_{k-1} \cdot 10^{k-1} + \cdots + d_1 \cdot 10 + d_0 \\ &= d_k \cdot (10^k - 1) + d_{k-1} \cdot (10^{k-1} - 1) + \cdots + d_1 \cdot (10 - 1) + d_0 \cdot (1 - 1) + (d_k + d_{k-1} + \cdots + d_1 + d_0) \\ &= d_k \cdot (10^k - 1) + d_{k-1} \cdot (10^{k-1} - 1) + \cdots + d_1 \cdot (10 - 1) + 3l. \end{aligned}$$

Now since $10^{j+1} - 1$ is divisible by 3 for every $j \geq 0$, therefore the quantity above is divisible by 3. □

Ex. 4.4.30.

- (a) Use the quotient-remainder theorem with $d = 3$ to prove that the product of any two consecutive integers has the form $3k$ or $3k + 2$ for some integer k .
 (b) Use the mod notation to rewrite the result of part (a).

SOLUTION . (a) Proof:

Suppose n and $n + 1$ are any two consecutive integers. By the quotient-remainder theorem with $d = 3$, we know that $n = 3q$, or $n = 3q + 1$, or $n = 3q + 2$ for some integer q .

Case 1: ($n = 3q$ for some integer q .)

In this case,

$$n(n + 1) = 3q(3q + 1) = 3[q(3q + 1)].$$

Let $k = q(3q + 1)$. Then k is an integer because sums and products of integers are integers.

Hence $n(n + 1) = 3k$ for some integer k .

Case 2: ($n = 3q + 1$ for some integer q .)

In this case,

$$n(n + 1) = (3q + 1)(3q + 2) = 9q^2 + 9q + 2 = 3(3q^2 + 3q) + 2.$$

Let $k = 3q^2 + 3q$. Then k is an integer because sums and products of integers are integers.

Hence $n(n + 1) = 3k + 2$ for some integer k .

Case 3: ($n = 3q + 2$ for some integer q .)

In this case,

$$n(n + 1) = (3q + 2)(3q + 3) = 3[(3q + 2)(q + 1)].$$

Let $k = (3q + 2)(q + 1)$. Then k is an integer because sums and products of integers are integers.

Hence $n(n + 1) = 3k$ for some integer k . □

- (b) Given any integer m , $m(m + 1) \bmod 3 = 0$ or $m(m + 1) \bmod 3 = 2$.

(Or equivalently, Given any integer m , $m(m + 1) \bmod 3 \neq 1$.)

Thus in all three cases, the product of the two consecutive integers either equals $3k$ or it equals $3k + 2$ for some integer k [as was to be shown]. □

Ex. 4.4.31. Prove that for all integers n and m , either $n - m$ and $m + n$ are both even, or they are both odd.**SOLUTION .** Fix arbitrary integers n and m . There are four cases.

Case 1: n is even and m is even: Let $n = 2k$ and $m = 2l$. Then $n - m = 2k - 2l = 2(k - l)$ and $n + m = 2k + 2l = 2(k + l)$ and so they are both even. By generalization, the statement “either $n - m$ and $m + n$ are both even, or they are both odd” is true.

Case 2: n is even and m is odd: Let $n = 2p$ and $m = 2q + 1$. Then $n - m = 2p - (2q + 1) = 2(p - q - 1) + 1$ and $n + m = 2p + (2q + 1) = 2(p + q) + 1$ and so they are both odd. By generalization, the statement “either $n - m$ and $m + n$ are both even, or they are both odd” is true.

Case 3: n is odd and m is even: This case is similar to Case 2.

Case 4: n is odd and m is odd: Let $n = 2r + 1$ and $m = 2s + 1$. Then $n - m = 2r - 2s = 2(r - s)$ and $n + m = 2r + 2s + 2 = 2(r + s + 1)$ and so they are both even. By generalization, the statement “either $n - m$ and $m + n$ are both even, or they are both odd” is true. □

Ex. 4.4.34. Given any integer n , if $n > 3$, could $n, n + 2$, and $n + 4$ all be primes? Prove or give a counterexample.

SOLUTION . Suppose n is any integer with $n > 3$. By the quotient-remainder theorem with $d = 3$, we know that $n = 3q$, or $n = 3q + 1$, or $n = 3q + 2$ for some integer q . Note that because n is greater than 3; q is greater than 1 or $q = 1$ and $n = 4$ or 5.

Case 1: ($q = 1$ and $n = 4$.)

In this case, n is not prime because $4 = 2 \cdot 2$.

Case 2: ($q = 1$ and $n = 5$.)

In this case, $n + 4 = 9 = 3 \cdot 3$, and so $n + 4$ is not prime.

Case 3: ($q > 1$ and $n = 3q$.)

In this case, n is not prime because it is a product of 3 and q and both 3 and q are greater than 1.

Case 4: ($q > 1$ and $n = 3q + 1$.)

In this case, $n + 2 = (3q + 1) + 2 = 3q + 3 = 3(q + 1)$. So $n + 2$ is not prime because it is a product of 3 and $q + 1$ and both 3 and $q + 1$ are greater than 1.

Case 5: ($q > 1$ and $n = 3q + 2$.)

In this case, $n + 4 = (3q + 2) + 4 = 3q + 6 = 3(q + 2)$. So $n + 4$ is not prime because it is a product of 3 and $q + 2$ and both 3 and $q + 2$ are greater than 1.

Hence in all five cases, at least one of n or $n + 2$ or $n + 4$ is not prime. □

Ex. 4.4.35. Prove that the fourth power of any integer has the form $8m$ or $8m + 1$ for some integer m .

SOLUTION . Let n be any integer. Then n is either even or odd.

Case 1: $n = 2k$ for some integer k . Then $n^4 = (2k)^4 = 16k^4 = 8(2k^4)$. So, n is of the form $8m$ where $m = 2k^4$.

Case 2: $n = 2l + 1$ for some integer l . Then $n^4 = (2l + 1)^4 = 16l^4 + 32l^3 + 24l^2 + 8l + 1 = 8(2l^4 + 4l^3 + 3l^2 + l) + 1$. So, n is of the form $8m + 1$, where $m = 2l^4 + 4l^3 + 3l^2 + l$. □

Ex. 4.4.50. If m, n , and d are integers, $d > 0$, and $d|(m - n)$, what is the relation between $m \bmod d$ and $n \bmod d$? Prove your answer.

SOLUTION . They are equal.

Proof: Suppose m, n and d are integers such that $d|(m - n)$. By definition of divisibility,

$$m - n = dk$$

for some integer k , and so

$$m = n + dk. \quad (*)$$

Let $r = n \bmod d$. Then by definition of $\bmod$,

$$n = qd + r, \quad (**)$$

where q and r are integers and $0 \leq r < d$. Therefore, we find that

$$\begin{aligned} m &= n + dk && \text{by } (*) \\ &= (qd + r) + dk && \text{by } (**) \\ &= d(q + k) + r && \text{by algebra.} \end{aligned}$$

Since $q + k$ is an integer and $0 \leq r < d$, the integral quotient of the division of m by d is $q + k$ and the remainder is r . Hence $m \bmod d = r$ also, and so $n \bmod d = m \bmod d$. □